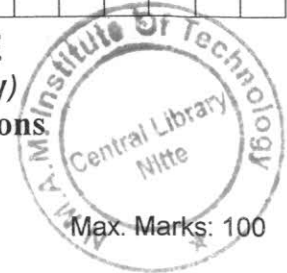


**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Second Semester B.Tech (CBCS) Degree Examinations**

May - June 2023

**EE1001-1 – BASIC ELECTRICAL ENGINEERING**



Duration: 3 Hours

**Note:**

*Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.*

*Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

**PART - A: MULTIPLE CHOICE QUESTIONS**

**20 Marks**

- In the case of nodal analysis, the equation at each node is written by applying \_\_\_\_\_  
 A) KVL  
 B) KCL  
 C) both KCL and KVL  
 D) None of these
- While calculating current using mesh analysis, it was found that the current in a particular branch containing  $4\ \Omega$  resistor is  $-2A$ . This means that \_\_\_\_\_  
 A) the  $4\ \Omega$  resistor is releasing (generating)  $2\ A$  current since the current is negative  
 B) the assumed direction of current in that resistor and the actual direction of flow of current are opposite to each other  
 C) the  $4\ \Omega$  resistor doesn't allow  $2A$  current to flow through it  
 D) none of these
- The analog electrical meters read the \_\_\_\_\_ of the quantity  
 A) Average value  
 B) RMS value  
 C) Instantaneous value  
 D) Maximum value
- Form factor is defined as \_\_\_\_\_ of the alternating quantity  
 A) Maximum value/RMS value  
 B) RMS value/Maximum value  
 C) RMS value/Average value  
 D) Average value/RMS value
- Read the following statements carefully:  
 i. Power factor is the ratio of reactive power to apparent power  
 ii. Power factor is the ratio of the resistance to the impedance of the circuit  
 iii. Power factor is the cosine of the angle between voltage and current  
 Now, out of the above statements, which statements are TRUE?  
 A) Statements i and ii  
 B) Statements ii and iii  
 C) Statements i and iii  
 D) All the three statements
- The power consumed by a pure resistor is \_\_\_\_\_  
 A) Reactive Power  
 B) Active Power  
 C) Apparent Power  
 D) None of these
- If a  $100V$ ,  $50Hz$ , single phase AC supplies a current of  $2A$  to a pure inductive circuit, the inductance of the circuit is \_\_\_\_\_  
 A)  $50\ H$   
 B)  $0.02\ H$   
 C)  $637\ mH$   
 D)  $0.159\ H$
- In a balanced three phase star circuit, neutral current is \_\_\_\_\_  
 A) Infinity  
 B) Zero  
 C) One  
 D) None of the above
- The function of the transformer is to \_\_\_\_\_  
 A) Convert AC to DC  
 B) Convert DC to AC  
 C) Step down or up the DC voltages & currents  
 D) Step down or up the AC voltages & currents
- A  $100V$ ,  $50Hz$  source is connected to the primary of a transformer having 20 turns. The maximum flux density in the core is  $1Wb/m^2$ . The cross-sectional area of core is \_\_\_\_\_  
 A)  $0.152\ m^2$   
 B)  $0.345\ m^2$   
 C)  $0.056\ m^2$   
 D)  $0.0225\ m^2$
- A single phase transformer has 400 primary and 1000 secondary turns. The net cross sectional area of the core is  $60\ cm^2$ . If the primary is connected to a  $500V$ ,  $50\ Hz$  source what is the voltage induced in the secondary?  
 A)  $950V$   
 B)  $1125V$   
 C)  $840V$   
 D)  $1250V$

12. Mutual inductance between two magnetically coupled coils depends on  
 A) Permeability of the core material B) Number of turns of the coil  
 C) Cross sectional area of their common core D) All of the above
13. Lap winding is suitable for \_\_\_\_\_ voltage d.c. generators applications.  
 A) Low B) High  
 C) Moderate D) Any
14. DC shunt motor is used in  
 A) Cranes B) Lathes  
 C) Hoists D) None of these
15. Which of the following part distinguishes a DC motor from an AC motor?  
 A) Winding B) Shaft  
 C) Commutator D) Stator
16. Which kind of rotor is most suitable for turbo alternators which are designed to run at high speed?  
 A) Salient pole type B) Non salient pole type  
 C) Both (A) and (B) D) None of the above
17. The types of rotors in 3 phase induction motor are  
 A) Salient pole & non-salient pole B) Salient pole & slip ring  
 C) Squirrel cage & slip ring D) Squirrel cage & smooth cylindrical
18. In a 3 Phase Induction motor, which type of the rotor winding is short circuited at both ends to two copper end rings  
 A) Squirrel cage B) Slip ring  
 C) Squirrel cage & slip ring D) Squirrel cage & phase wound
19. The fuse material must have low  
 A) conductivity B) melting point  
 C) permittivity D) none of these
20. The resistance of earthing electrode should be  
 A) Low B) high  
 C) medium D) very high

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

#### Unit – I

- |                                                                                                                                                                                                                                                                                                                                                          | Marks | BT* | CO* | PO* |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) Define the rms value and obtain the expression for rms value of a sinusoidal alternating current.                                                                                                                                                                                                                                                  | 5     | L2  | 1   | 1   |
| b) An alternating voltage is varying sinusoidally with a frequency of 50 Hz and has an rms value of 230 V. Its instantaneous value at $t=0$ is zero.<br>(i) Write down the equation for the instantaneous value of the voltage.<br>(ii) Find the instantaneous value at 0.125s.<br>(iii) At what time instant is the instantaneous value equal to 230 V? | 5     | L2  | 2   | 2   |
| c) In Fig. Q1(c), find the voltage across $8\Omega$ resistor using mesh analysis. Also find out the power consumed by the said resistor.                                                                                                                                                                                                                 |       |     |     |     |

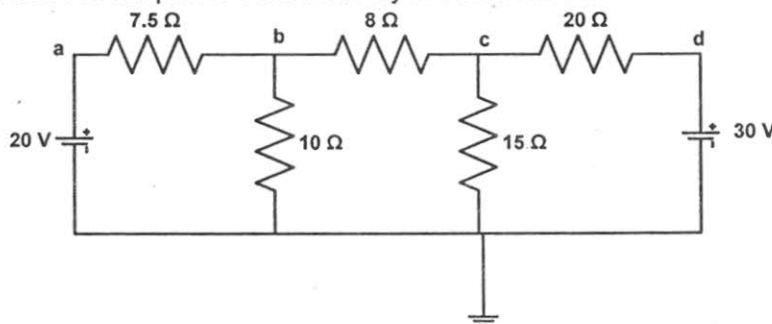


Fig. Q1(c)

- |                                                                                                                                                     |   |    |   |     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|-----|
| 2. a) Analyze a series AC R-L circuit with waveforms and phasor diagram. Hence obtain an expression for the average power consumed by this circuit. | 6 | L4 | 2 | 1,2 |
| b) Define: (i) Form factor (ii) Peak factor<br>(iii) Average Value (iv) Frequency of an alternating quantity.                                       | 5 | L2 | 2 | 2   |
| c) With appropriate diagrams, prove that the line voltage in a star-connected 3-phase AC system is $\sqrt{3}$ times the phase voltage.              | 5 | L3 | 2 | 1   |

3. a) With relevant equations show that the power consumed by a balanced three-phase load can be measured by two wattmeters. 7 L3 2 1
- b) Two wattmeters connected to a 3-phase motor indicate the total power input to be 12kW. The power factor is 0.6. Determine the readings of each wattmeter. 5 L2 2 2
- c) With a neat figure, describe the power triangle. 4 L1 1 1
- Unit – II**
4. a) Derive the emf equation of a transformer. 5 L3 3 1
- b) Derive the expression for the armature torque of a DC motor. 5 L3 4 1
- c) A 16-pole star-connected alternator has 144 slots and 10 conductors per slot. The flux per pole is 30 mWb and the speed is 375 rpm. Find the frequency, the phase emf and line emf. Assume that the winding is concentrated and full-pitched ( $K_d = K_p = 1$ ). 6 L2 4 2
5. a) Prove that the efficiency of a transformer is maximum when its Copper loss equals its iron loss. 6 L3 3 1,2
- b) Discuss the three characteristics of a DC shunt motor. 5 L2 4 1
- c) Describe the working principle of a 3-phase synchronous motor. 5 L2 4 1
6. a) Obtain the expression for the frequency of the induced emf of an alternator in terms of number of poles and rpm. 4 L3 4 1
- b) A 4-pole DC generator has a lap-wound armature with 50 slots with 16 conductors per slot. The useful flux per pole is 30 mWb. Determine the speed at which the machine must be driven to generate an e.m.f. of 240 V. 6 L2 4 2
- c) A 250-kVA, single-phase transformer has an efficiency of 96 % on full load at 0.8 power factor lagging and also on half load 0.8 power factor lagging. Find its iron loss and full-load copper loss. 6 L2 3 2
- Unit – III**
7. a) With neat figures, elaborate the production of a rotating magnetic field in a three-phase induction motor. 6 L2 5 1
- b) Comment why a squirrel cage induction motor is the most popular among the motors. 4 L2 5 1
- c) A 4-pole, 3-phase induction motor operates from a supply whose frequency is 50Hz. Calculate:
- (i) The speed at which the magnetic field of the stator is rotating.
- (ii) The speed of the rotor when the slip is 0.04.
- (iii) Frequency of the rotor current for the above slip of 0.04. 6 L2 5 2
8. a) Discuss conduit wiring and list its advantages and disadvantages. 5 L2 5 1
- b) Write a comparative note on fuse and MCB. 6 L1 5 1
- c) Describe pipe earthing with a neat figure. 5 L2 5 1

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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